

Lecture 24/Module 3

Sinusoidal Steady State Analysis

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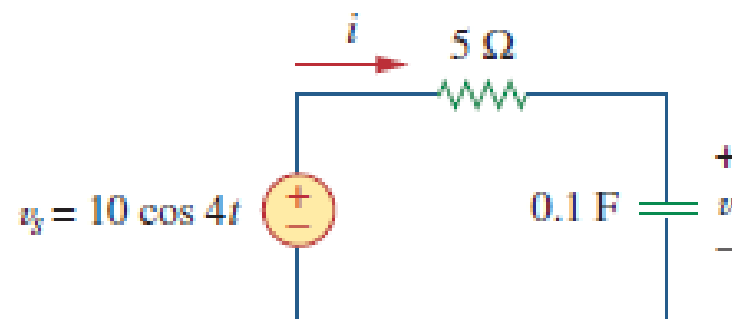
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Steps to Analyze AC Circuits:

1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.

Exercise -1 (Voltage Division)

Find $v(t)$ and $i(t)$ in the circuit shown



Answer

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

$i(t)$ leads $v(t)$ by 90° as expected

Solution

Find $v(t)$ and $i(t)$ in the circuit shown

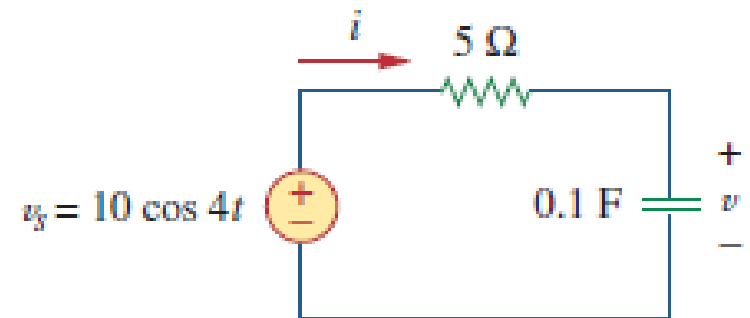
Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$



Hence the current

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10\angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789\angle 26.57^\circ \text{ A}\end{aligned}$$

The voltage across the capacitor is

$$\begin{aligned}\mathbf{V} &= \mathbf{I}\mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789\angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789\angle 26.57^\circ}{0.4\angle 90^\circ} = 4.47\angle -63.43^\circ \text{ V}\end{aligned}$$

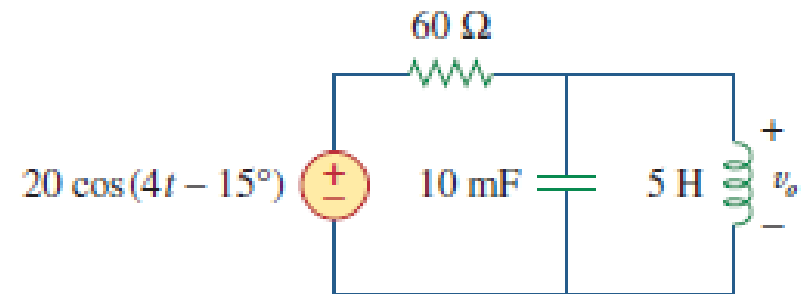
$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

$i(t)$ leads $v(t)$ by 90° as expected

Exercise -2 (Voltage Division)

Determine $v_o(t)$ in the circuit



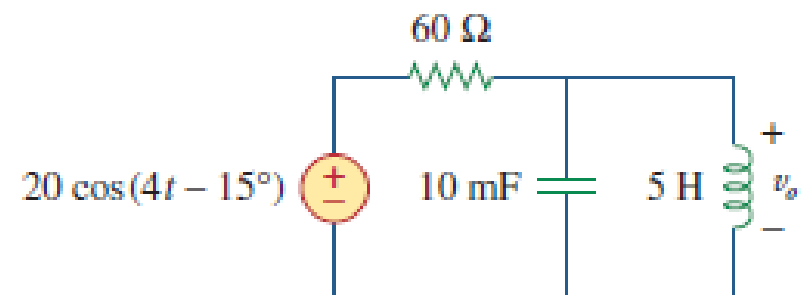
Answer

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \, \text{V}$$

Solution

Determine $v_o(t)$ in the circuit

Solution:



$$v_s = 20 \cos(4t - 15^\circ) \quad \Rightarrow \quad \mathbf{V}_s = 20 \angle -15^\circ \text{ V}, \quad \omega = 4$$

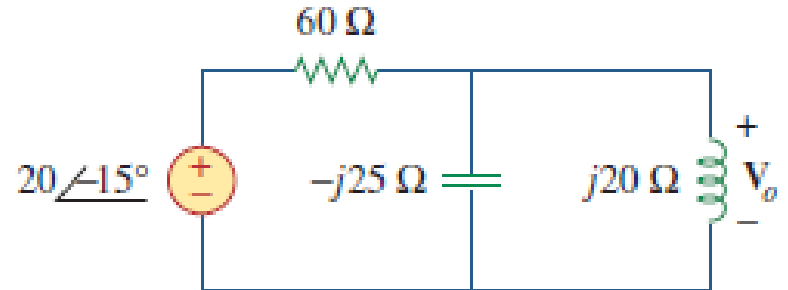
$$10 \text{ mF} \quad \Rightarrow \quad \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} \\ = -j25 \Omega$$

$$5 \text{ H} \quad \Rightarrow \quad j\omega L = j4 \times 5 = j20 \Omega$$

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \text{ V}$$

Z_1 = Impedance of the 60- Ω resistor

Z_2 = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor



$$Z_1 = 60\ \Omega$$

$$Z_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100\ \Omega$$

By the voltage-division principle,

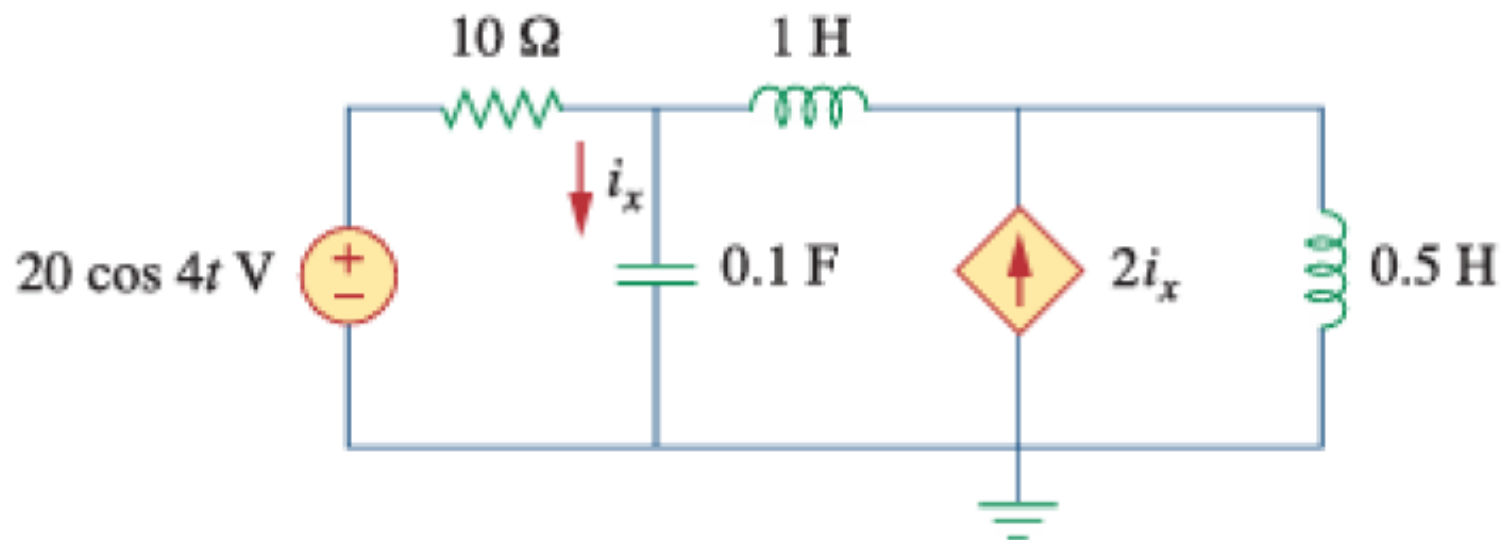
$$\begin{aligned} V_o &= \frac{Z_2}{Z_1 + Z_2} V_s = \frac{j100}{60 + j100} (20\angle -15^\circ) \\ &= (0.8575\angle 30.96^\circ) (20\angle -15^\circ) = 17.15\angle 15.96^\circ\ \text{V} \end{aligned}$$

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ)\ \text{V}$$

The frequency domain equivalent of the circuit

Exercise -3 (Node Analysis)

Find I_x in the circuit using nodal analysis method.



Answer

The current I_x is given by

$$I_x = \frac{V_1}{-j2.5} = \frac{18.97 \angle 18.43^\circ}{2.5 \angle -90^\circ} = 7.59 \angle 108.4^\circ \text{ A}$$

Transforming this to the time domain,

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

Solution

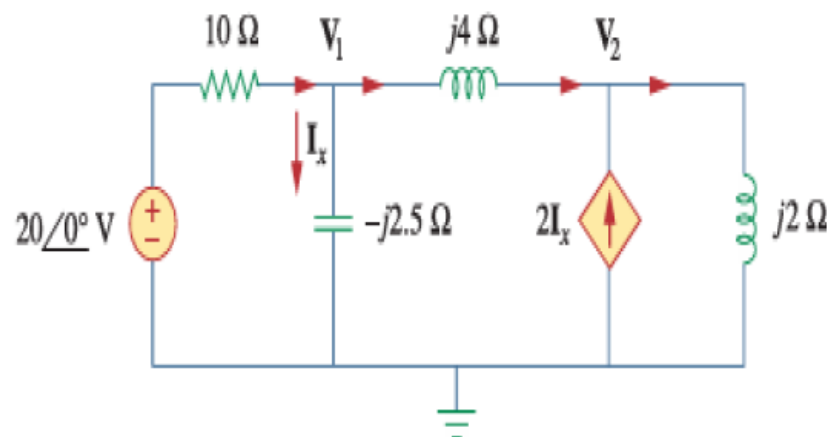
We first convert the circuit to the frequency domain:

$$20 \cos 4t \Rightarrow 20 \angle 0^\circ, \quad \omega = 4 \text{ rad/s}$$

$$1 \text{ H} \Rightarrow j\omega L = j4$$

$$0.5 \text{ H} \Rightarrow j\omega L = j2$$

$$0.1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j2.5$$



Applying KCL at node 1,

$$\frac{20 - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4}$$

or

$$(1 + j1.5)\mathbf{V}_1 + j2.5\mathbf{V}_2 = 20$$

At node 2,

$$2\mathbf{I}_x + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

But $\mathbf{I}_x = \mathbf{V}_1 / -j2.5$. Substituting this gives

$$\frac{2\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

By simplifying, we get

$$11\mathbf{V}_1 + 15\mathbf{V}_2 = 0$$

$$\mathbf{V}_1 = \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97 \angle 18.43^\circ \text{ V}$$

$$\mathbf{V}_2 = \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91 \angle 198.3^\circ \text{ V}$$

The current $\mathbf{I}_x$ is given by

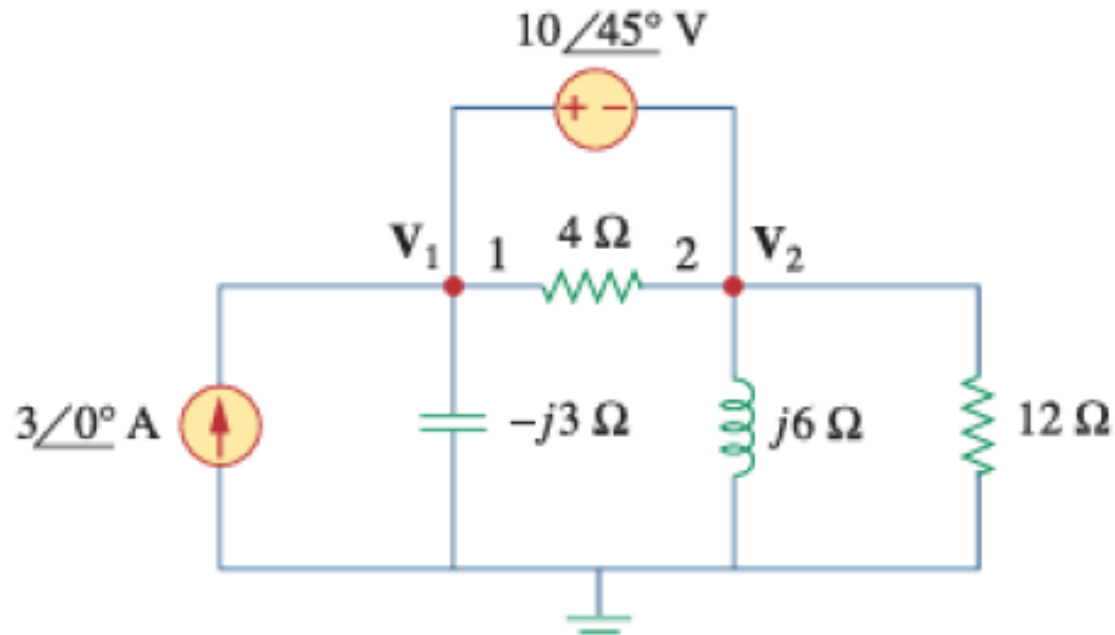
$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j2.5} = \frac{18.97 \angle 18.43^\circ}{2.5 \angle -90^\circ} = 7.59 \angle 108.4^\circ \text{ A}$$

Transforming this to the time domain,

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

Exercise -4 (Node Analysis-Super Node)

Compute V_1 and V_2 in the circuit using nodal analysis method.

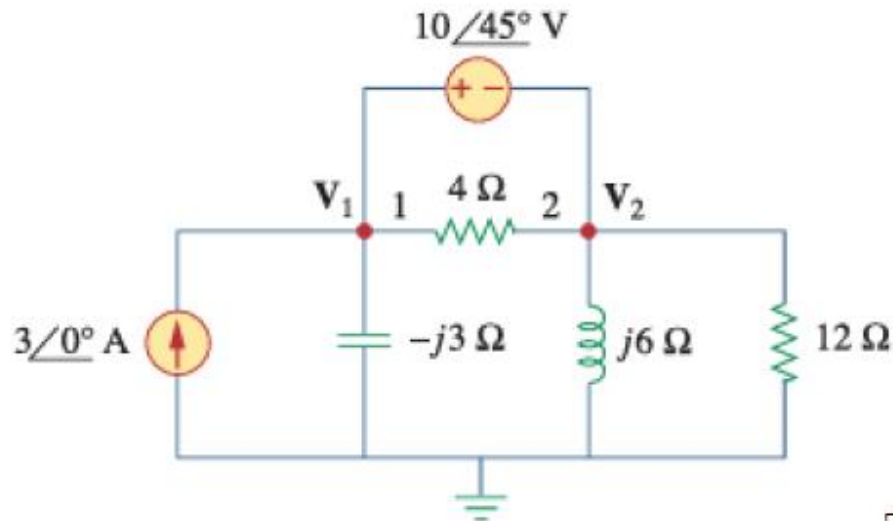


Answer

$$V_2 = 31.41\angle -87.18^\circ \text{ V}$$

$$V_1 = V_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$

. Compute V_1 and V_2 in the circuit using nodal analysis method.



Applying KCL at supernode gives,

$$3 = \frac{V_1}{-j3} + \frac{V_2}{j6} + \frac{V_2}{12}$$

$$36 = j4V_1 + (1 - j2)V_2 \quad \text{---1}$$

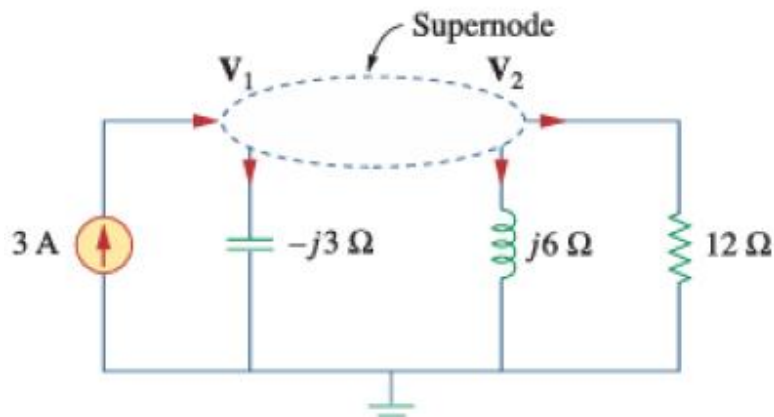
The voltage source is between nodes V_1 and V_2

$$V_1 = V_2 + 10\angle45^\circ \quad \text{---2}$$

Substituting eqn.2 in eqn.1 gives,

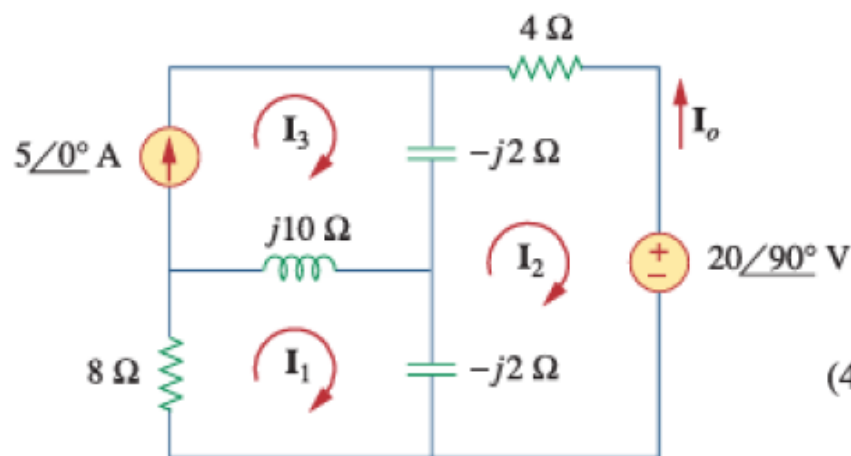
$$36 - 40\angle135^\circ = (1 + j2)V_2 \Rightarrow V_2 = 31.41\angle-87.18^\circ \text{ V}$$

$$V_1 = V_2 + 10\angle45^\circ = 25.78\angle-70.48^\circ \text{ V}$$



Exercise -5 (Mesh Analysis)

1. Compute the current I_o in the circuit using mesh analysis method.



Applying KVL to all the meshes gives,

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0 \quad \text{---1}$$

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0 \quad \text{---2}$$

From mesh 3, I_3 is 5 A, hence, eqn.1 and eqn.2 becomes,.

$$(8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 = j50$$

$$j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 = -j20 - j10$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{416.17\angle -35.22^\circ}{68} = 6.12\angle -35.22^\circ \text{ A}$$

$$\mathbf{I}_o = -\mathbf{I}_2 = 6.12\angle 144.78^\circ \text{ A}$$

Thank You